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$$I = \int \frac{1}{1 - \sin x} dx$$

$$\text{Stel } t = \tan\left(\frac{x}{2}\right)$$

$$\sin x = \frac{2t}{1+t^2} \Rightarrow dx = \frac{2}{1+t^2} dt$$

Dus:

$$\begin{aligned} I &= \int \frac{1}{1 - \frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt = \int \frac{1+t^2}{1+t^2-2t} \cdot \frac{2}{1+t^2} dt \\ &= \int \frac{1+t^2}{(t-1)^2} \cdot \frac{2}{1+t^2} dt = \int \frac{2}{(t-1)^2} dt \\ &= 2 \int (t-1)^{-2} dt = 2 \cdot \frac{(t-1)^{-1}}{-1} + c = \frac{-2}{t-1} + c \end{aligned}$$

Terug substitueren:

$$I = \frac{-2}{\tan\left(\frac{x}{2}\right) - 1} + c$$

References